DIGITAL FORENSICS

AppleTV Activity Analysis

Surveillance Case: Ann & Mr. X

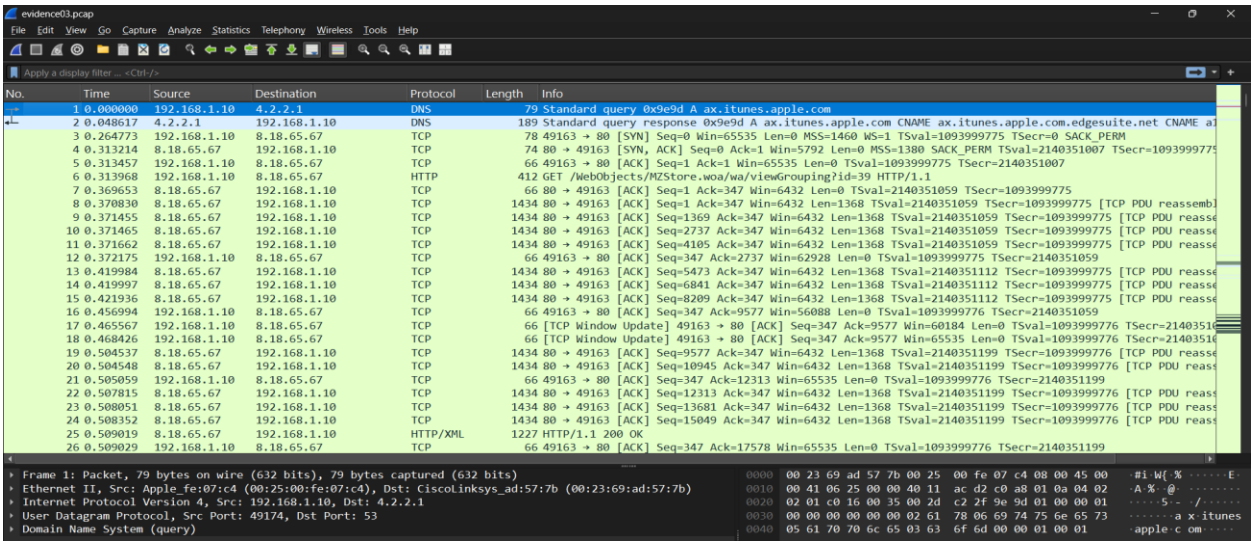
File: Evidence03.pcapng | Tool: Wireshark

Field	Detail
Case File	evidence03.pcap
Subject	Ann (suspect) — AppleTV device
Device IP	192.168.1.10 (static)
Analysis Tool	Wireshark 4
Protocol Focus	DNS, TCP, HTTP

Section 1 — Case Background & Scenario

Ann and Mr. X have established a new base of operations while extradition paperwork is processed. As part of a covert surveillance operation, investigators have captured all network traffic generated by Ann's AppleTV device (192.168.1.10). This report documents a forensic analysis of that capture file, reconstructing Ann's browsing behaviour, search activity, and media interests from the raw packet data.

The AppleTV communicates with Apple's iTunes Store servers over unencrypted HTTP, meaning all search queries, movie selections, and server responses are fully readable within the capture. This represents a significant intelligence opportunity.



Section 2 — Device Identification

Network Participants

IP Address	Device / Role	Notes
192.168.1.10	Ann's AppleTV (subject device)	Static IP, Apple OUI confirmed
4.2.2.1	DNS Resolver	External public DNS server
8.18.65.x	Apple Akamai CDN servers	iTunes Store content delivery
66.235.132.x	Apple metrics servers	Analytics / telemetry
224.0.0.251	Multicast (mDNS)	Apple Bonjour service discovery

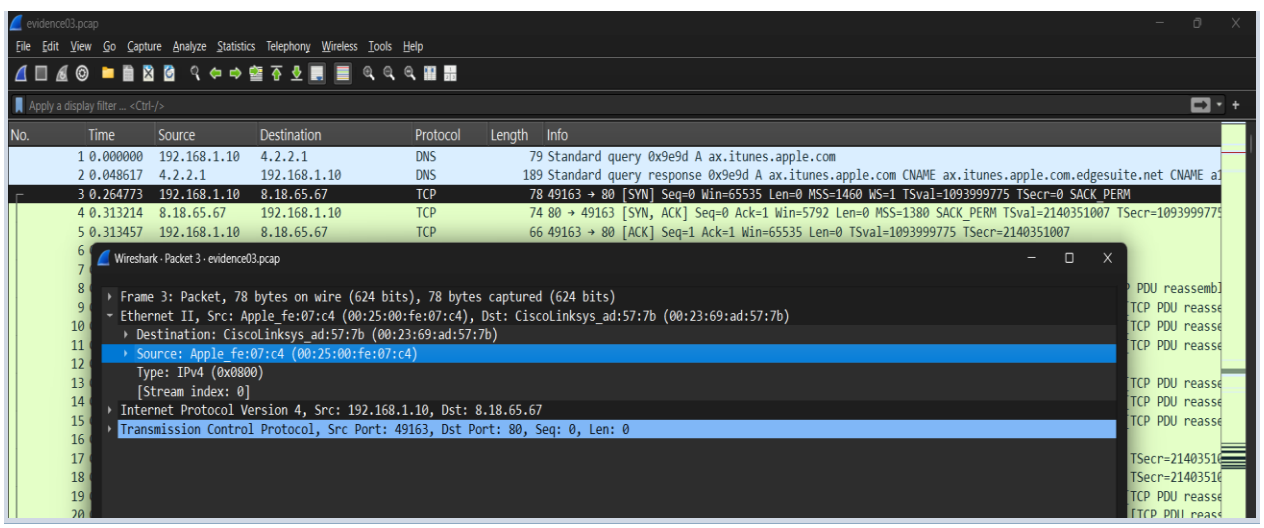
Section 3 — Forensic Questions & Answers

Q1. What is the MAC address of Ann's AppleTV?

Every packet originating from 192.168.1.10 carries the same source MAC address in the Ethernet header. This was confirmed by filtering on the device's IP and inspecting the Layer 2 frame fields.

The OUI prefix 00:25:00 is registered to Apple Inc., which is consistent with an Apple hardware device. The device also self-identifies as "AppleTV.local" in mDNS (Multicast DNS) broadcast packets (e.g. Packet 36).

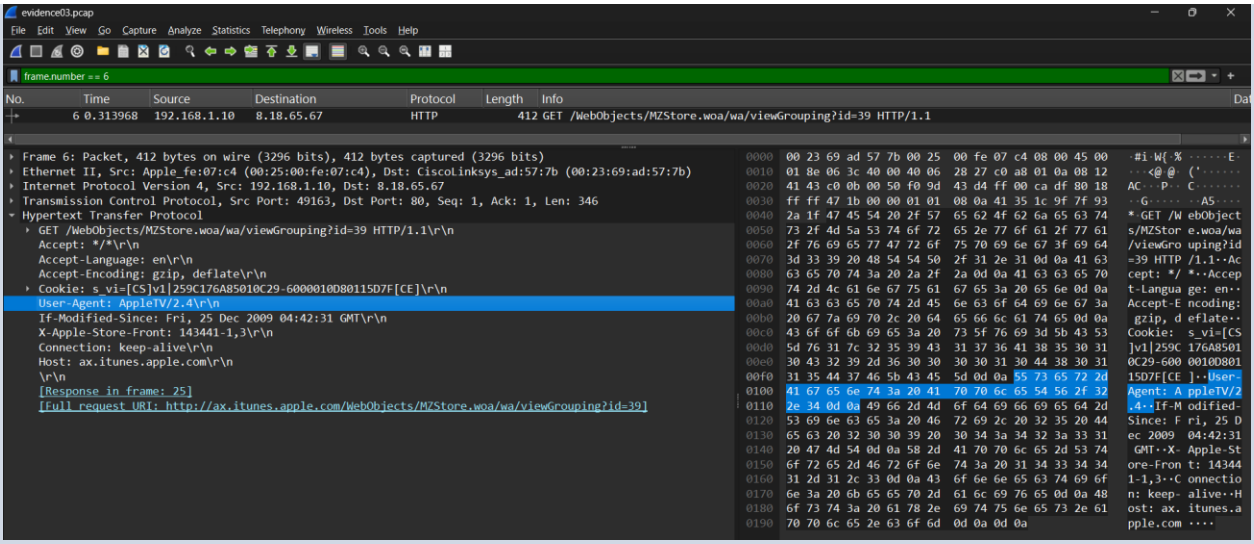
MAC ADDRESS	`00:25:00:fe:07:c4`



Q2. What User-Agent string did Ann's AppleTV use in HTTP requests?

The User-Agent header is sent by the device with every HTTP request, identifying itself to the server. All HTTP requests from 192.168.1.10 carry the same User-Agent string, which confirms the device model and software version.

USER-AGENT	`AppleTV/2.4`



Q3. What were Ann's first four search terms on the AppleTV? (all incremental searches count)

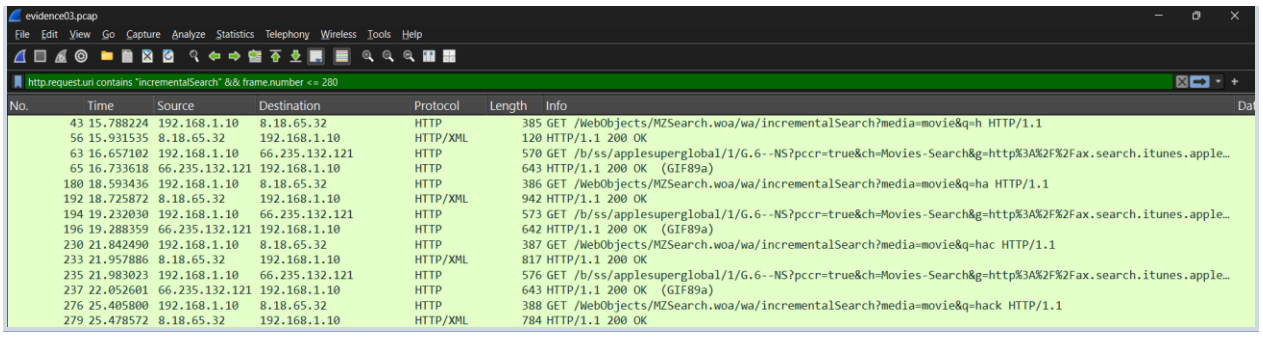
The AppleTV sends a new HTTP GET request to ax.search.itunes.apple.com after every single keystroke typed into the search bar. This 'incremental search' behaviour allows the store to suggest results in real time - and also exposes every letter Ann typed to network inspection.

The search terms, in order of transmission, were:

#	Search Term	Packet	Wireshark URI (q= parameter)
1	h	43	incrementalSearch?media=movie&q=h
2	ha	180	incrementalSearch?media=movie&q=ha
3	hac	230	incrementalSearch?media=movie&q=hac
4	hack	276	incrementalSearch?media=movie&q=hack

Ann was typing the word "hack" - indicating an interest in hacking-themed movies on the iTunes Store.

SEARCH TERMS	`h → ha → hac → hack`



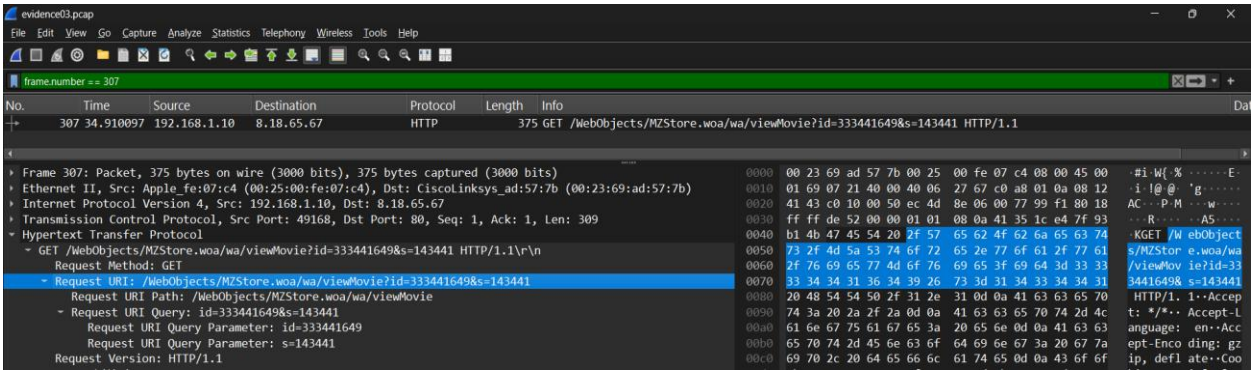
Q4. What was the title of the first movie Ann clicked on?

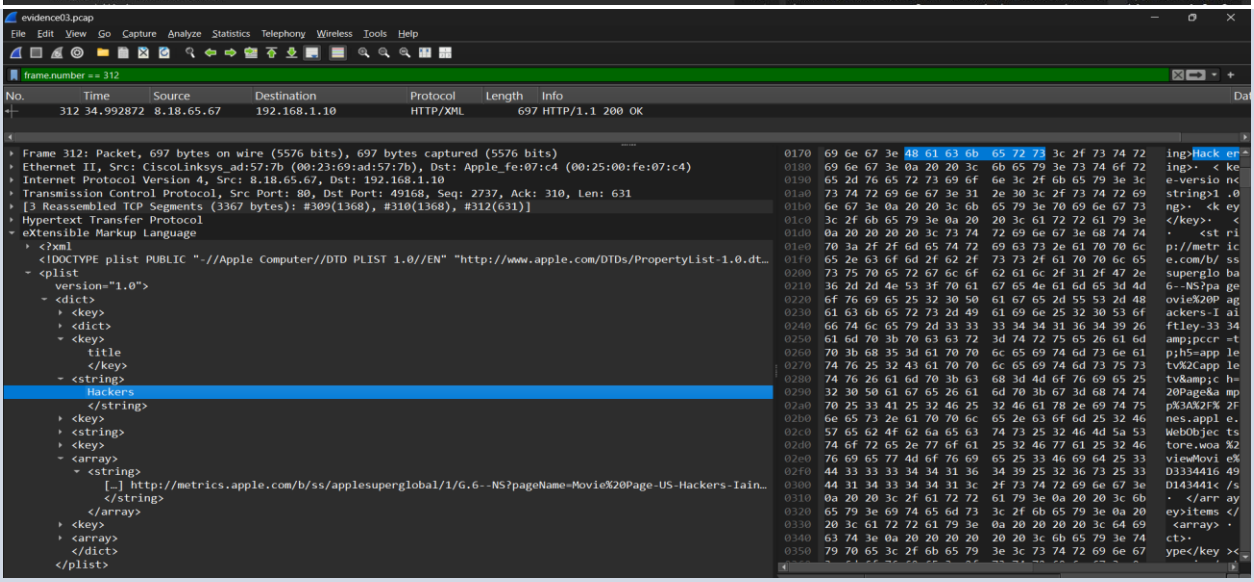
After completing her 'hack' search, Ann selected a movie result. At Packet 307, the AppleTV sends a GET request to the iTunes Store for a specific movie ID (333441649). The server's XML response at Packet 312 contains the full movie metadata, including the title field.

The metrics ping URL in Packet 320 also confirms the selection:

```
pageName=Movie%20Page-US-Hackers-Iain%20Softley-333441649
```

MOVIE TITLE	Hackers (1995, dir. Iain Softley)



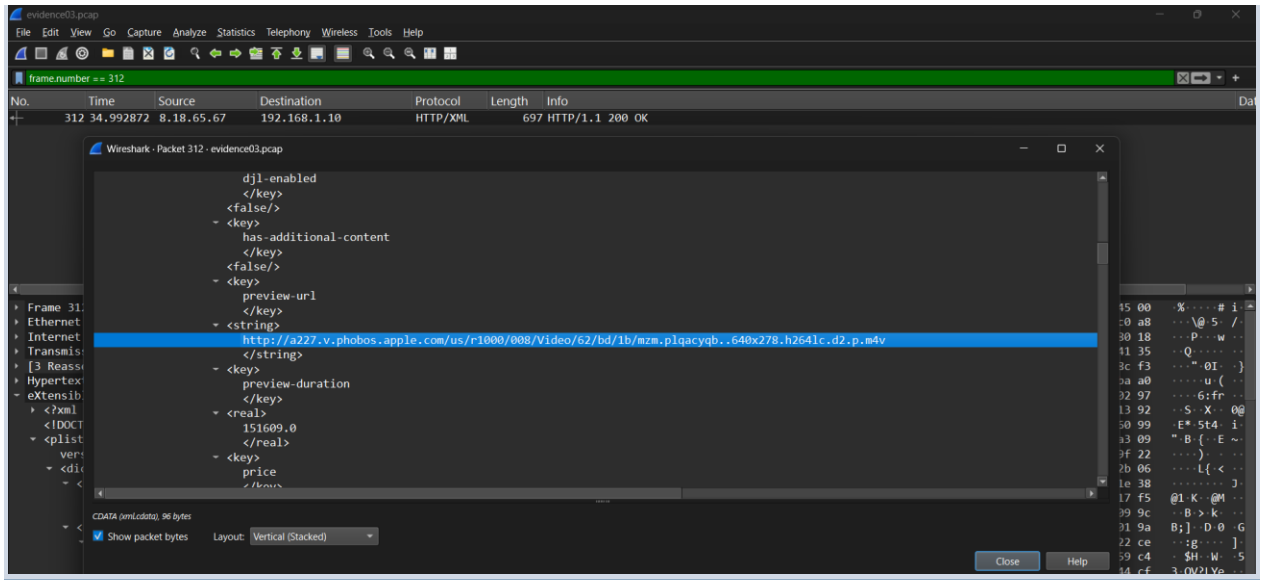


Q5. What was the full URL to the movie trailer (defined by "preview-url")?

The iTunes Store XML response for the Hackers movie page (Packet 312) contains a 'preview-url' field for each purchase/rental tier. All tiers point to the same trailer file hosted on Apple's Phobos media servers.

PREVIEW-URL	`http://a227.v.phobos.apple.com/us/r1000/008/Video/62/bd/1b/mzm.plqacyqb..640x2`

The filename extension .m4v and path structure (640x278, h264lc) indicate an H.264 widescreen trailer at 640×278 resolution.



Q6. What was the title of the second movie Ann clicked on?

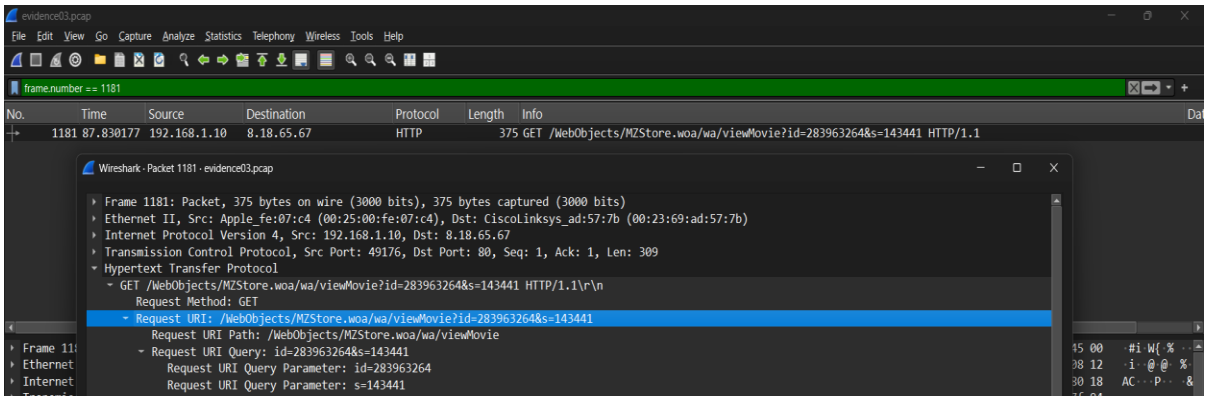
After viewing the Hackers page, Ann started a new search — typing 's → sn → sne → sneb → snea → sneak'. She then clicked on a second movie. At Packet 1181, the AppleTV fetches movie ID 283963264. The server's XML response at Packet 1186 reveals the title.

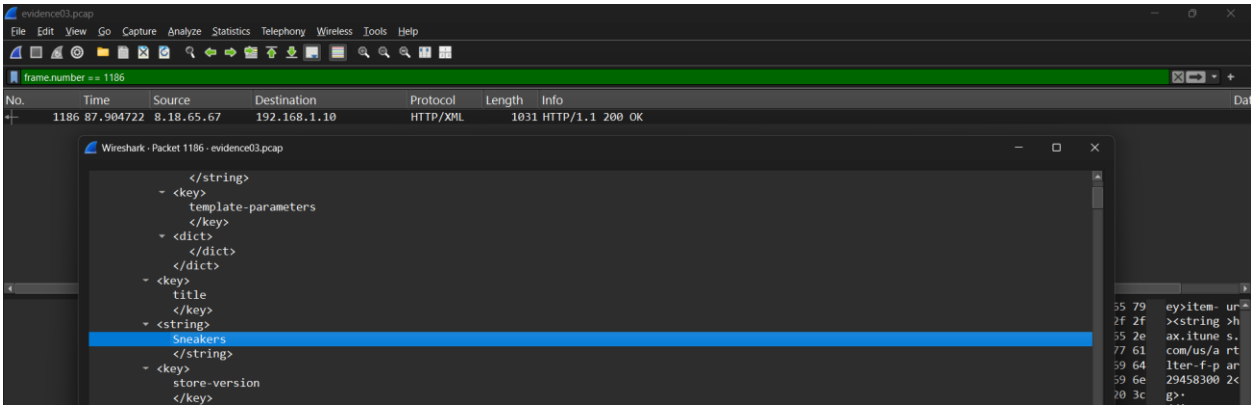
The metrics URL at Packet 1189 further confirms:

```
pageName=Movie%20Page-US-Sneakers-Phil%20Alden%20Robinson-283963264
```

MOVIE TITLE	`Sneakers (1992, dir. Phil Alden Robinson)`

Rated PG-13. Genre: Action & Adventure. 43 user ratings averaging 0.88/1.0. Copyright 2008 Universal Studios.

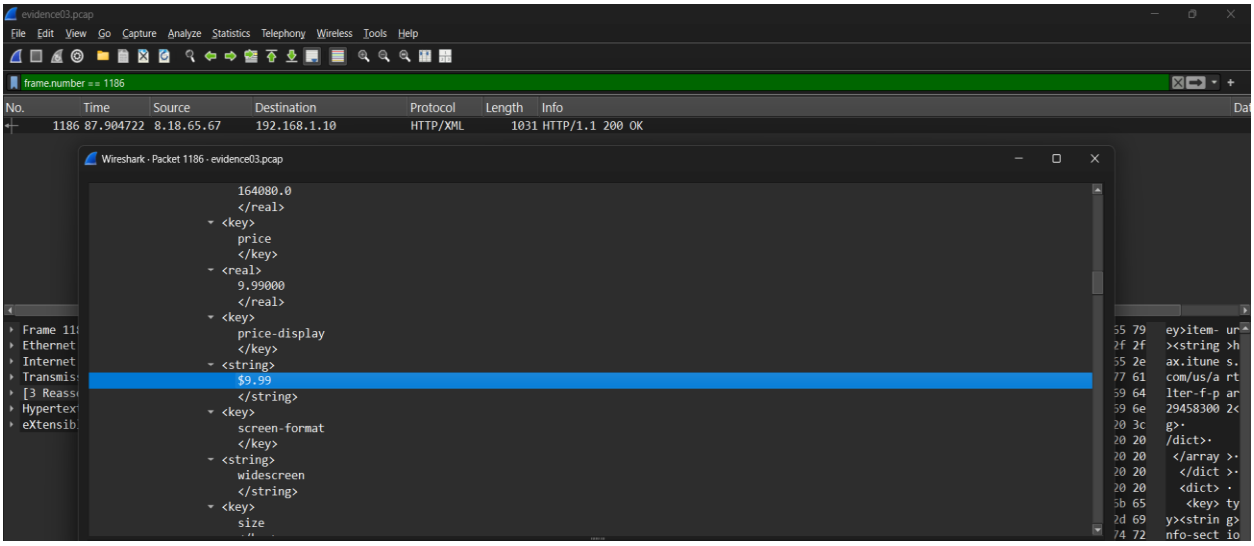




Q7. What was the price to buy it (defined by "price-display")?

The iTunes Store XML response for Sneakers (Packet 1186) contains store-offer pricing data for each available purchase tier. The 'price-display' field gives the human-readable price as shown to the customer on-screen.

PRICE-DISPLAY	`$9.99 (Standard Definition buy price)`



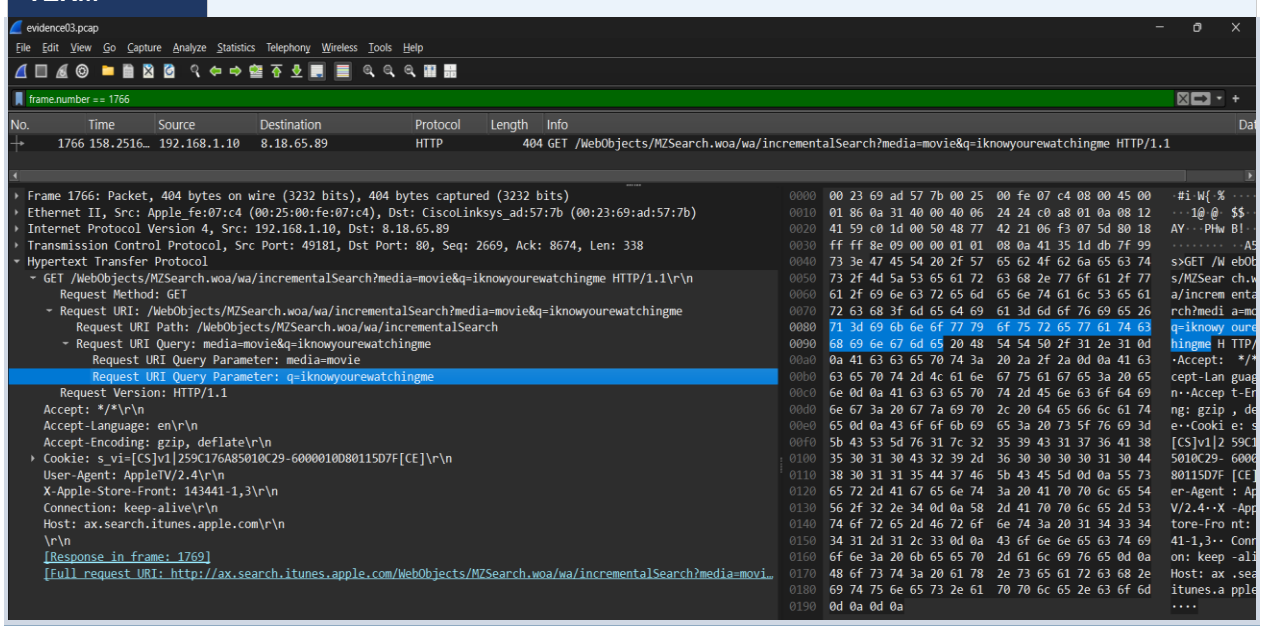
Q8. What was the last full term Ann searched for?

After browsing Sneakers, Ann began a completely new incremental search, this time starting with 'i'. She typed the full phrase character by character. The final complete term, the last search query captured in the PCAP, was sent at Packet 1766.

The full incremental sequence was:

Step	Query so far	Step	Query so far
1	i	10	iknowyour
2	ik	11	iknowyoure
3	ikn	12	iknowyourew
4	ikno	13	iknowyourewa
5	iknow	14	iknowyourewat
6	iknowy	15	iknowyourewatc
7	iknowyo	16	iknowyourewatch
8	iknowyou	17	iknowyourewatchi
9	iknowyour	18	iknowyourewatchin
—	—	19	iknowyourewatching
—	—	20	iknowyourewatchingm
—	—	21	iknowyourewatchingme

LAST SEARCH TERM `iknowyourewatchingme`



Section 4 — Suspect Activity Profile

Based on the complete packet capture analysis, the following activity profile was constructed for Ann:

Activity	Detail	Forensic Significance
Device	AppleTV 2nd gen (v2.4), MAC 00:25:00:fe:07:c4, IP 192.168.1.10	Confirms physical device identity
First Search	Typed 'hack' incrementally — browsed Hackers (1995)	Interest in hacking culture / cybercrime films
Second Search	Typed 'sneak' — browsed Sneakers (1992)	Interest in surveillance & intelligence themes
Third Search	Full phrase: 'iknowyourewatchingme'	Awareness of or response to covert monitoring
Trailer Viewed	Hackers trailer via Phobos CDN (preview-url recovered)	Confirms active movie browsing, not passive
Price Checked	Sneakers buy price $9.99 retrieved from iTunes Store	Possible intent to purchase

Section 5 — Protocol Summary

Protocol	Role in This Capture	Key Packets
DNS	Resolves iTunes Store hostnames (ax.itunes.apple.com, ax.search.itunes.apple.com)	1–2, 38–39, 58–60, 866–867 …
TCP	Reliable transport layer — 3-way handshakes before every HTTP session	3–5, 40–42, 304–306 …
HTTP	All iTunes Store requests (plaintext — fully readable)	6, 43, 116, 180, 230, 276, 307, 1181, 1766 …
mDNS	Apple Bonjour service discovery (AppleTV announces itself on LAN)	35, 36, 284, 301, 1469–1470
ARP	Layer 2 address resolution — reveals device MAC address	2, 3

Section 6 — Key Forensic Findings

1. HTTP traffic from the AppleTV is entirely unencrypted — every search query, movie selection and server response is visible in the packet capture without any decryption required.
2. Incremental (keystroke-by-keystroke) search requests leak the user's full intended search term even if they never pressed 'Search' — each character is transmitted individually.
3. The iTunes Store XML responses expose rich metadata: movie titles, prices, ratings, trailer URLs, and purchase parameters — all recoverable from a single packet.
4. The device's MAC address (OUI: Apple Inc.) and mDNS service announcements allow positive identification of the device make and model even without any login credentials.
5. Ann's final search phrase — 'iknowyourewatchingme' — may constitute evidence of counter-surveillance awareness and should be treated as a significant intelligence indicator.
6. No HTTPS/TLS was used for iTunes Store communications in this capture, which is consistent with the older AppleTV 2.4 software. Newer Apple devices encrypt all store traffic.

This report was produced for educational and GitHub portfolio purposes. All analysis performed on a provided PCAP file using Wireshark. No real criminal investigation is implied.